PREDICTION OF PROTEIN-DRUG INTERACTIONS

Fundamentals of Data Science

Sapienza University of Rome

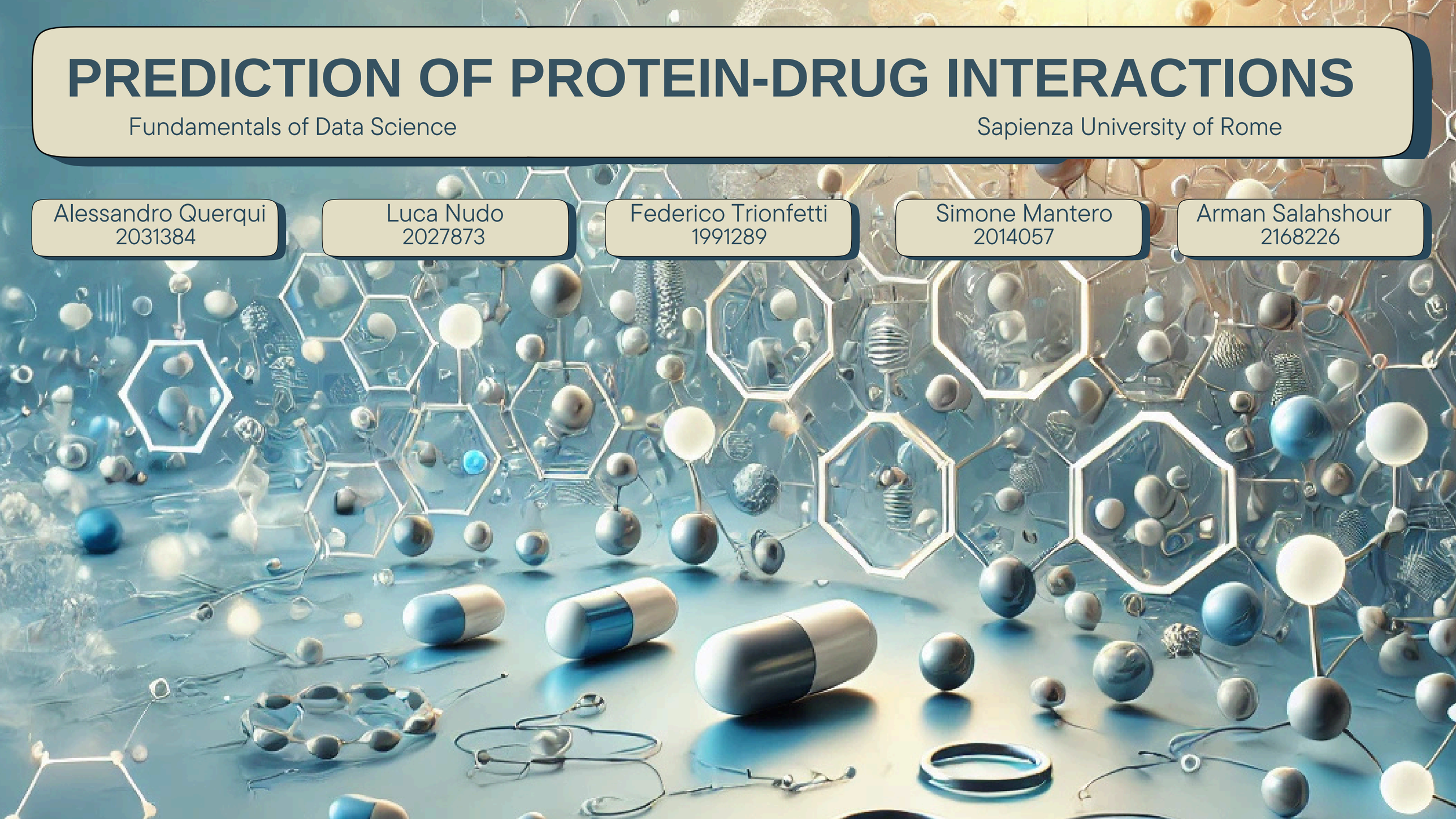
Alessandro Querqui
2031384

Luca Nudo
2027873

Federico Trionfetti
1991289

Simone Mantero
2014057

Arman Salahshour
2168226



Understanding Drug-Protein Interactions (DPIs)

- DPIs are essential to drug discovery and understanding therapeutic mechanisms.
- Traditional DPI research methods rely heavily on laboratory experiments:
  - Time-consuming
  - Resource-intensive
- Machine learning (ML) provides a scalable, efficient alternative.

Study Objectives

- The machine learning model aims to predict the likelihood of interactions between the given drug-like molecules and their protein targets, improving the efficacy and safety of therapeutic drug development.

- Through the use of state-of-the-art algorithms and feature engineering techniques, this study seeks to provide insights into the molecular mechanisms underlying DPIs, facilitating the identification of novel drug candidates with higher precision and reduced risk of adverse effects.

Dataset Description

The dataset has dimensions 47.190 x 5. It includes:

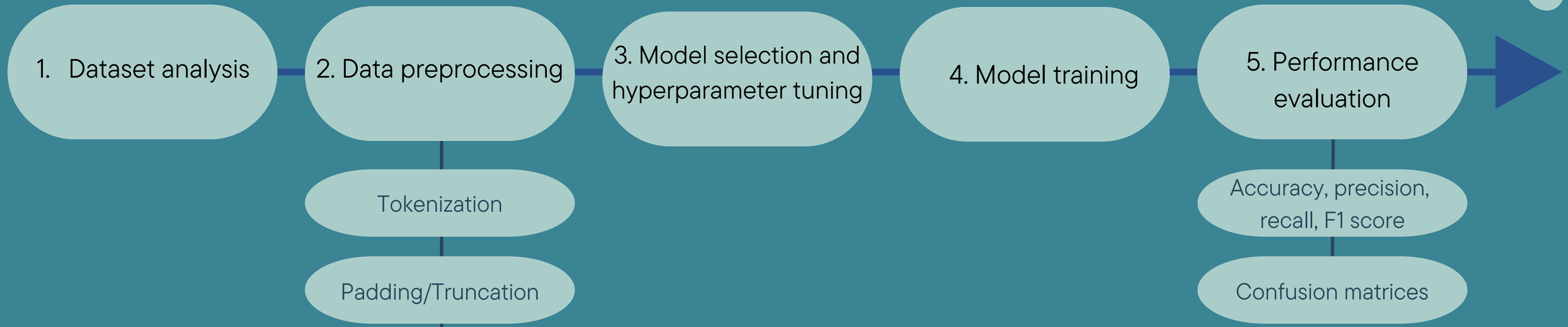
- Epitope Sequences: epitope amino acid (aa) and SMILES representations (smi)
- T-cell Receptors: Names (tcr) and chemical structures (tcr_full).
- Labels: Binary values indicating interaction (1) or non-interaction (0)

	epitope_aa	epitope_smi	tcr	tcr_full	label
0	FLKEKGGL	CC(C)C[C@H](NC(=O)CNC(=O)CNC(=O)[C@H](CCCCN)NC...	CSVWGTGKTYEQYF	SAVISQKPSRDICQRGTSLTIQCQVDSQVTMMFWYRQQPGQSLTLI...	1
1	FLKEKGGL	CC(C)C[C@H](NC(=O)CNC(=O)CNC(=O)[C@H](CCCCN)NC...	CSVWGEGRSYEQYF	SAVISQKPSRDICQRGTSLTIQCQVDSQVTMMFWYRQQPGQSLTLI...	1
2	FLKEKGGL	CC(C)C[C@H](NC(=O)CNC(=O)CNC(=O)[C@H](CCCCN)NC...	CSATILAGVPYGEQYF	GAVVSQHPSWVICKSGTSVKIECRSLDFQATTMFWYRQFPKQSLML...	1
3	FLKEKGGL	CC(C)C[C@H](NC(=O)CNC(=O)CNC(=O)[C@H](CCCCN)NC...	CSASEGTSSYEQYF	GAVVSQHPSWVICKSGTSVKIECRSLDFQATTMFWYRQFPKQSLML...	1
4	FLKEKGGL	CC(C)C[C@H](NC(=O)CNC(=O)CNC(=O)[C@H](CCCCN)NC...	CASSFDREVTGELFF	GAGVSQTPSNKVTEKGKYVELRCDPISGHTALYWYRQSLGQGPEFL...	1

Subsequently tokenization, padding, integer mapping and feature engineering were carried out to prepare it for use.

The dataset was split in train:70%, validation:10%, test:20%, with 330.330 training examples.

Pipeline overview



Comparison of 3 machine learning models

Random Forest	• Ensemble of decision trees for robust, interpretable predictions: handles structured data well but lower precision. • Parameters: n_estimators=100, max_depth=10, min_samples_split=5
XGBoost	• Gradient boosting for improved performance with.sequential tree building: it excels with structured and high-dimensional data. • Parameters: max_depth=6, learning_rate=0.1, subsample=0.8.
Convolutional Neural Networks (CNNs)	• Specialized in extracting patterns from sequence and structural data; higher computational complexity. • 4 convolutional layers, increasing filters (64–128), kernel size (3,1).

Performance Metrics



- **CNN**: High recall, suitable for reducing false negatives
- **XGBoost**: Best overall performance (accuracy and precision)
- **Random Forest**: Lower performance but reliable recall

Confusion Matrices Overview

CNN:

	Predicted Negative	Predicted Positive
Actual Negative	2535	1028
Actual Positive	512	3004

Trade-off between precision and recall

XGBoost:

	Predicted Negative	Predicted Positive
Actual Negative	3578	1166
Actual Positive	1033	3661

Good performance, but slightly reduced recall compared to CNN.

Random Forest:

	Predicted Negative	Predicted Positive
Actual Negative	3468	1276
Actual Positive	1169	3525

Good recall, but higher number of false positives compared to the other models.

Conclusions

This study compared Random Forest, XGBoost, and Convolutional Neural Networks (CNN) for predicting drug-protein interactions, evaluating their performance using accuracy, precision, recall, F1 score, and confusion matrices.

XGBoost emerged as the best-performing model with the highest accuracy (0.7834) and precision (0.7584), while CNN demonstrated superior recall (0.8544). Random Forest showed lower performance overall, with an accuracy of 0.7409 and precision of 0.7342.

Despite CNN's strengths in handling sequential data, its high computational requirements make XGBoost a more practical choice for large-scale datasets. XGBoost strikes an effective balance between accuracy, precision, and computational efficiency, making it the most suitable model for this task.

Future improvements could involve refining hyperparameters and incorporating richer data features to enhance predictive performance.

Future Directions

Although this study showed good results, there are several potential avenues for improvement and further exploration:

- Integration of Pre-trained Models: leverage transfer learning for improved accuracy
- Incorporation of Richer Features: 3D molecular structures, protein-ligand binding sites
- Use of Graph Neural Networks (GNNs): Advanced architectures for biological data
- Ensemble Approaches: Combine strengths of different models for robust predictions

Thank you for your attention!
Feel free to ask questions

Link to the dataset: https://tdcommons.ai/multi_pred_tasks/tcrepitope/

Link to Therapeutics Data Commons webpage: https://tdcommons.ai/

Link to Github repository having some useful codes on this dataset:
https://github.com/Arman-Salahshour/Protein-to-Drug-Prediction-with-LLM

Link to Stanfield RL, Wilson IA. Protein-peptide interactions. 1995 Feb:
https://pubmed.ncbi.nlm.nih.gov/7773739/